LAB NO. 06

INHERITANCE IN C++

(HIERARCHICAL, MULTIPLE, HYBRID)

Lab Objectives

Following are the lab objectives:

1. Understand the concept of inheritance in Object-Oriented Programming.
2. Learn how to create base and derived classes in C++.
3. Implement types of inheritance in programs.
4. Apply inheritance to promote code reusability and better program structure.

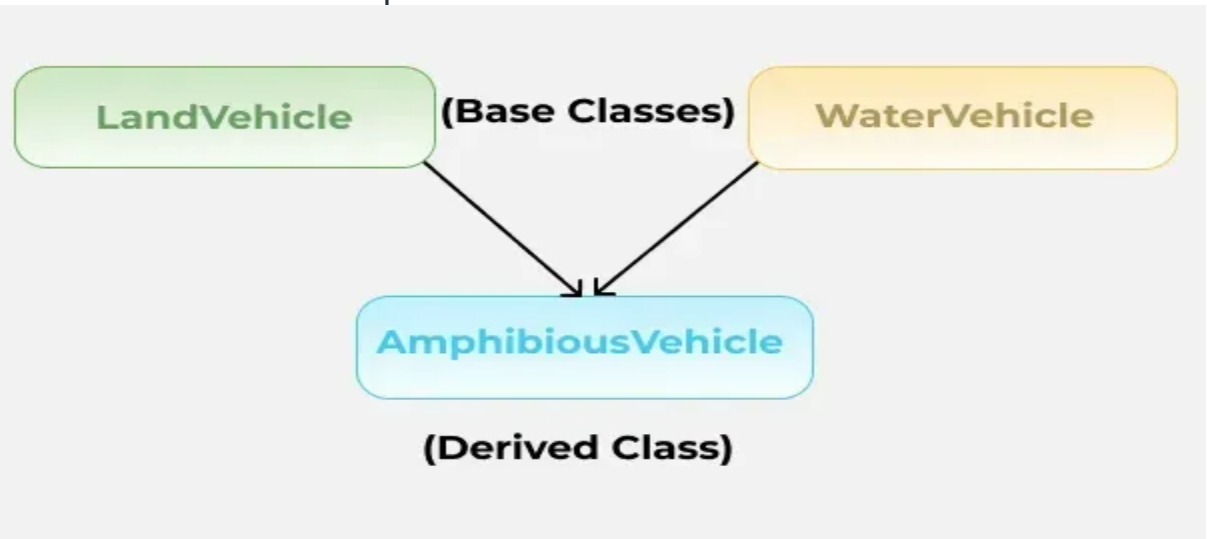
Student ID	**9864**	**Student Name**	**Ali Raza**

	Marks	**Obtained Marks**	**Comments**
Task 1	**20**		
Task 2	**20**		
Task 3	**20**		
Total Marks	**60**		

Lab Instructor

Multiple Inheritance

In Multiple inheritance, one class can have more than one superclass and inherit features from all parent classes.



Example:

```
#include <iostream>
using namespace std;

class LandVehicle
{
  public:
    void landInfo()
    {
        cout << "This is a LandVehicle" << endl;
    }
};

class WaterVehicle
{
  public:
    void waterInfo()
    {
        cout << "This is a WaterVehicle" << endl;
    }
};
```

```cpp
// Derived class inheriting from both base classes
class AmphibiousVehicle : public LandVehicle, public WaterVehicle
{
  public:
    AmphibiousVehicle()
    {
        cout << "This is an AmphibiousVehicle" << endl;
    }
};

int main()
{
    AmphibiousVehicle obj;

    obj.waterInfo();
    obj.landInfo();

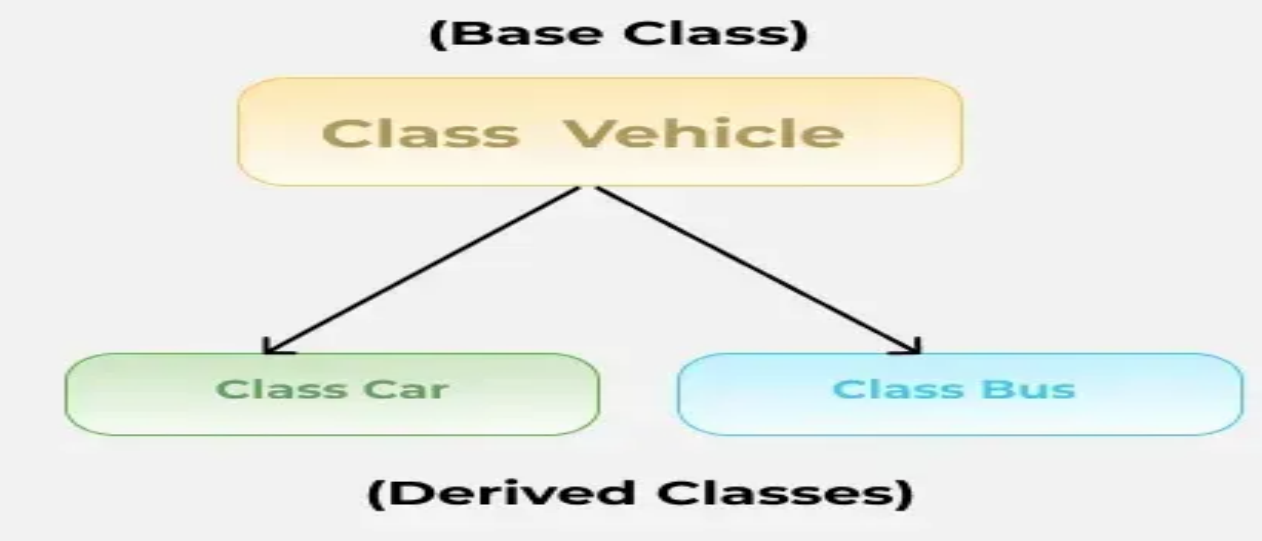
    return 0;
}
```

Output

```
This is an AmphibiousVehicle
This is a WaterVehicle
This is a LandVehicle
```

Hierarchical Inheritance

In hierarchical inheritance, more than one subclass is inherited from a single base class. i.e. more than one derived class is created from a single base class. For example, cars and buses both are vehicle.



Example:

```
#include <iostream>
using namespace std;

class Vehicle
{
  public:
    Vehicle()
    {
        cout << "This is a Vehicle" << endl;
    }
};

class Car : public Vehicle
{
  public:
    Car()
    {
        cout << "This Vehicle is Car" << endl;
    }
};
```

```
class Bus : public Vehicle
{
  public:
    Bus()
    {
        cout << "This Vehicle is Bus" << endl;
    }
};

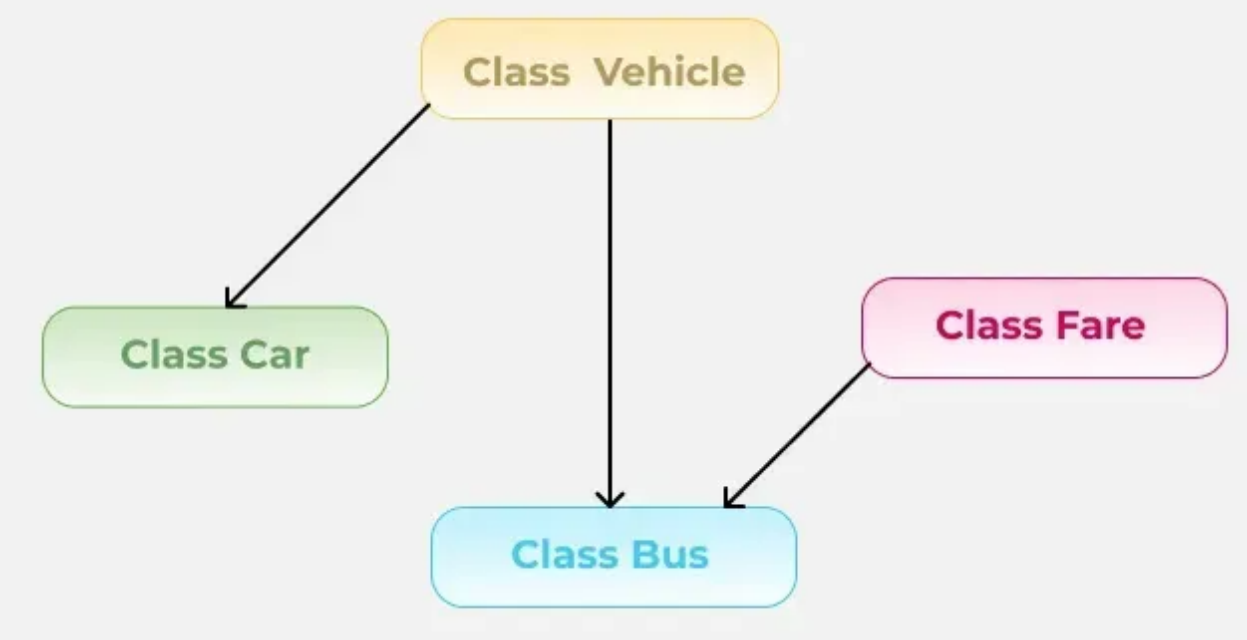
int main()
{
    Car obj1;
    Bus obj2;
    return 0;
}
```

Output

```
This is a Vehicle
This Vehicle is Car
This is a Vehicle
This Vehicle is Bus
```

Hybrid Inheritance

When two or more types of inheritance are combined in one program. For example, a class might use multiple inheritance and also be part of a multilevel inheritance chain.



Example:

```
#include <iostream>
using namespace std;

class Vehicle
{
  public:
    Vehicle()
    {
        cout << "This is a Vehicle" << endl;
    }
};

class Fare
{
  public:
    Fare()
    {
        cout << "Fare of Vehicle" << endl;
    }
};
```

```
class Car : public Vehicle
{
  public:
    Car()
    {
        cout << "This Vehical is a Car" << endl;
    }
};

class Bus : public Vehicle, public Fare
{
  public:
    Bus()
    {
        cout << "This Vehicle is a Bus with Fare";
    }
};

int main()
{
    Bus obj2;
    return 0;
}
```

Output

```
This is a Vehicle
Fare of Vehicle
This Vehicle is a Bus with Fare
```

Note:

Hybrid Inheritance can lead to the diamond problem in C++. This happens when a class inherits from two classes that both share the same base class. As a result the derived class gets multiple copies of the base class members, which creates ambiguity about which one to use.

> ***Note :*** *The solution is to use virtual inheritance, so only a single copy of the base class is shared.*

LAB TASKS

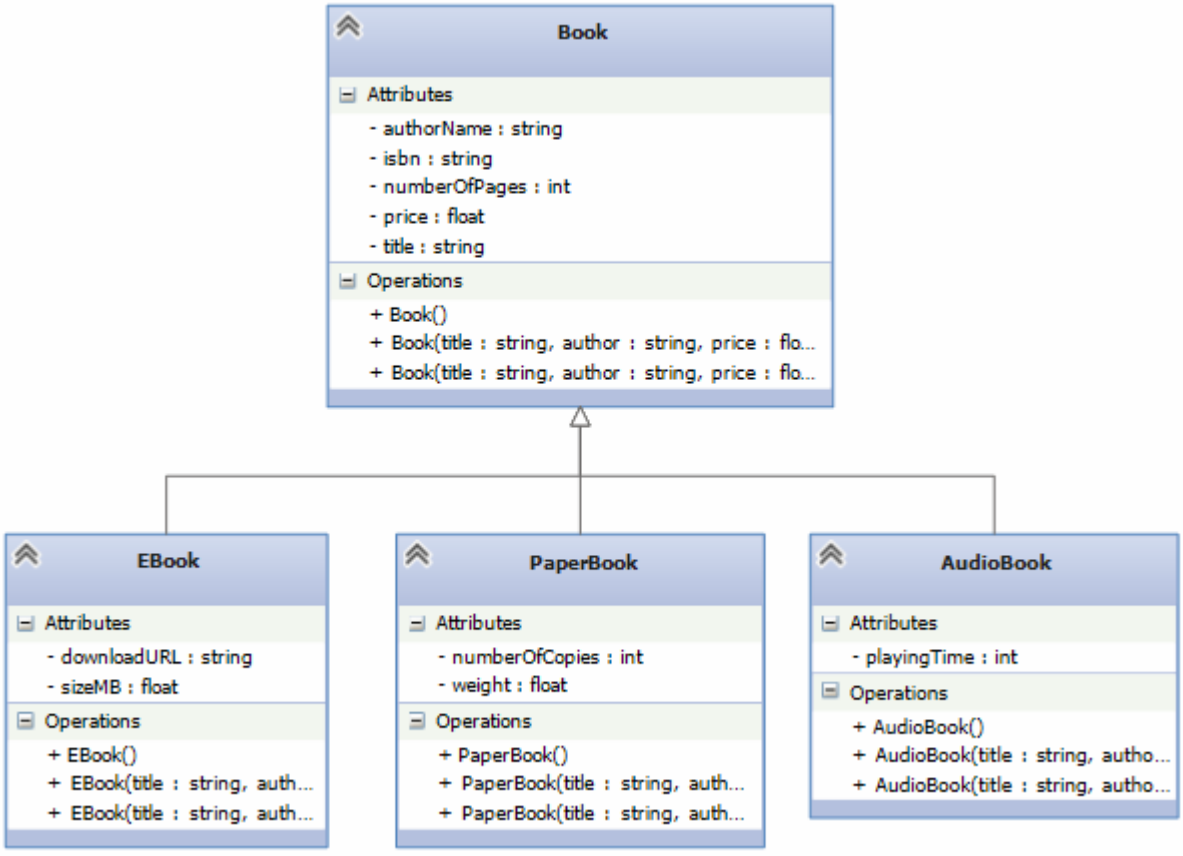
Hierarchical Inheritance:

Task 1: Book, EBook, AudioBook, PaperBook (20 mark)

See the following class diagram of **Book**, **ElectronicBook**, **AudioBook**, & **PaperBook** and implement inheritance relationship of classes in python.

You need to implement following items for each class;

1. Constructors
2. Setters
3. Getters
4. Member Methods



Program:

```cpp
#include <iostream>
#include <string>
using namespace std;
class Book {
private:
    string authorName;
    string bookTitle;
    int numPages;
    double price;
public:
    Book() {
        authorName = "not given ";
       bookTitle = "x";
        numPages = 0;
     price = 0.0;
    }
    Book(string title, string author, int pages, double p) {
      bookTitle = title;
      authorName = author;
      numPages = pages;
       price = p;
    }
       void setBookDetails(string title, string author, int pages, double p) {
     bookTitle = title;
      authorName = author;
       numPages = pages;
         price = p;
    }
    void displayBookInfo() {
      cout << "Title:" << bookTitle << endl;
      cout << "Author:" << authorName << endl;
      cout << "Pages:" << numPages << endl;
       cout << "Price of book in Pakistani rupees is:   " << price << endl;
```

```
    }
};
class ElectronicBook : public Book {
private:
    int numberOfCopies;
    double  weight;

public:
    ElectronicBook() {
        numberOfCopies = 0;
        weight = 0.0;
    }
    void setElectronicBookDetails(string title, string author, int pages, double p, int copies, double w) {
     setBookDetails(title, author, pages, p);
    numberOfCopies = copies;
       weight = w;
    }

    void displayEBookInfo() {
        displayBookInfo();
     cout << "Copies:" << numberOfCopies << endl;
       cout << "Weight:" << weight << endl;
    }
};
class PaperBook : public Book {
private:
    int numOfCopies;
    double weight;

public:
    PaperBook() {
        numOfCopies = 0;
        weight = 0.0;
```

```
    }

    void setPaperBookDetails(string title, string author, int pages, double p, int copies, double w) {
        setBookDetails(title, author, pages, p);
      numOfCopies = copies;
       weight = w;
    }

    void displayPaperBookInfo() {
        displayBookInfo();
     cout << "Copies: " << numOfCopies << endl;
       cout << "Weight: " << weight << "" << endl;
    }
};
class AudioBook : public Book {
private:
    int playingTime;
    string sName;



public:
    AudioBook() {
        playingTime = 0;
        sName = "not given";
    }

    void setAudioBookDetails(string title, string author, int pages, double p, int time, string speaker) {
        setBookDetails(title, author, pages, p);
        playingTime = time;
        sName = speaker;
    }

    void displayAudioBookInfo() {
```

```
        displayBookInfo();
        cout << "Playing Time: " << playingTime << endl;
        cout << "Speaker: " << sName << endl;
    }
};

int main() {
    ElectronicBook e;
    e.setElectronicBookDetails("Rich dad poor dad ", "Dan Brown", 88, 9., 100, 2.5);
    e.displayEBookInfo();

    cout << endl;
    cout << endl;

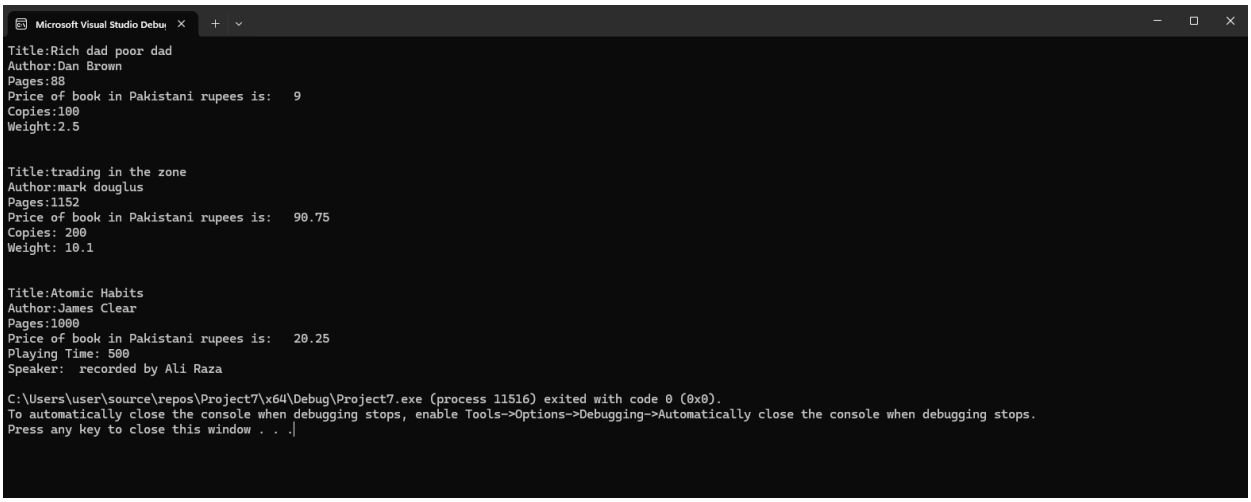
    PaperBook p;
    p.setPaperBookDetails("trading in the zone ", "mark douglus", 1152, 90.75, 200, 10.10);
    p.displayPaperBookInfo();

    cout << endl;
    cout << endl;


    AudioBook a;
            a.setAudioBookDetails("Atomic Habits", "James Clear", 1000, 20.25, 500, " recorded by Ali Raza ");
    a.displayAudioBookInfo();

    return 0;
}
```

Output:


Microsoft Visual Studio Debu
Title:Rich dad poor dad
Author:Dan Brown
Pages:88
Price of book in Pakistani rupees is: 9
Copies:100
Weight:2.5
Title:trading in the zone
Author:mark douglus
Pages:1152
Price of book in Pakistani rupees is: 90.75
Copies: 200
Weight: 10.1
Title:Atomic Habits
Author:James Clear
Pages:1000
Price of book in Pakistani rupees is: 20.25
Playing Time: 500
Speaker: recorded by Ali Raza
C:\Users\user\source\repos\Project7\x64\Debug\Project7.exe (process 11516) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .

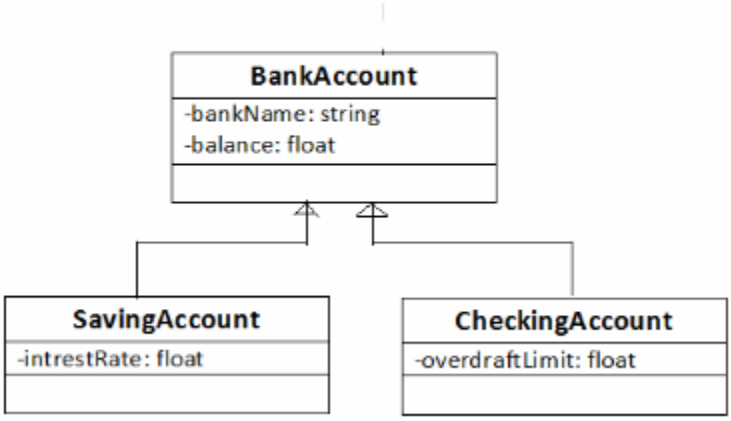
Task 2: Multiple Inheritance (20 mark)

See the following class diagram and implement multi-level inheritance relationship of classes in C++.

You need to implement following items for each class;

1. Constructors
2. Setters
3. Getters
4. Member Methods

Demonstrate working of your classes.



Program:

```
#include <iostream>
using namespace std;
class BankAccount;

class SavingAccount {
private:
  double interestRate;

public:
 SavingAccount() {
    interestRate = 0.0;
```

```
    }
    void setSavingDetails(double rate) {
        interestRate = rate;
    }
    double getInterestRate() {
        return interestRate;
    }

    void displaySavingInfo() {
    cout << "Interest Rate: " << interestRate << "%" << endl;
    }
};
class CheckingAccount : public SavingAccount {
private:
    double overdrawLimit;

public:
    CheckingAccount() {
    overdrawLimit = 0.0;
    }
    void setCheckingDetails(double rate, double limit) {
      setSavingDetails(rate);
        overdrawLimit = limit;
    }

    double getOverdrawLimit() {
        return overdrawLimit;
    }

    void displayCheckingInfo() {
          displaySavingInfo();
      cout << "Overdraw Limit: Rs" << overdrawLimit << endl;
    }
```

```
};
class BankAccount : public CheckingAccount {
private:
    string bankName;
    double balance;

public:
    BankAccount() {
    bankName = "not given/nill";
      balance = 0.0;
    }

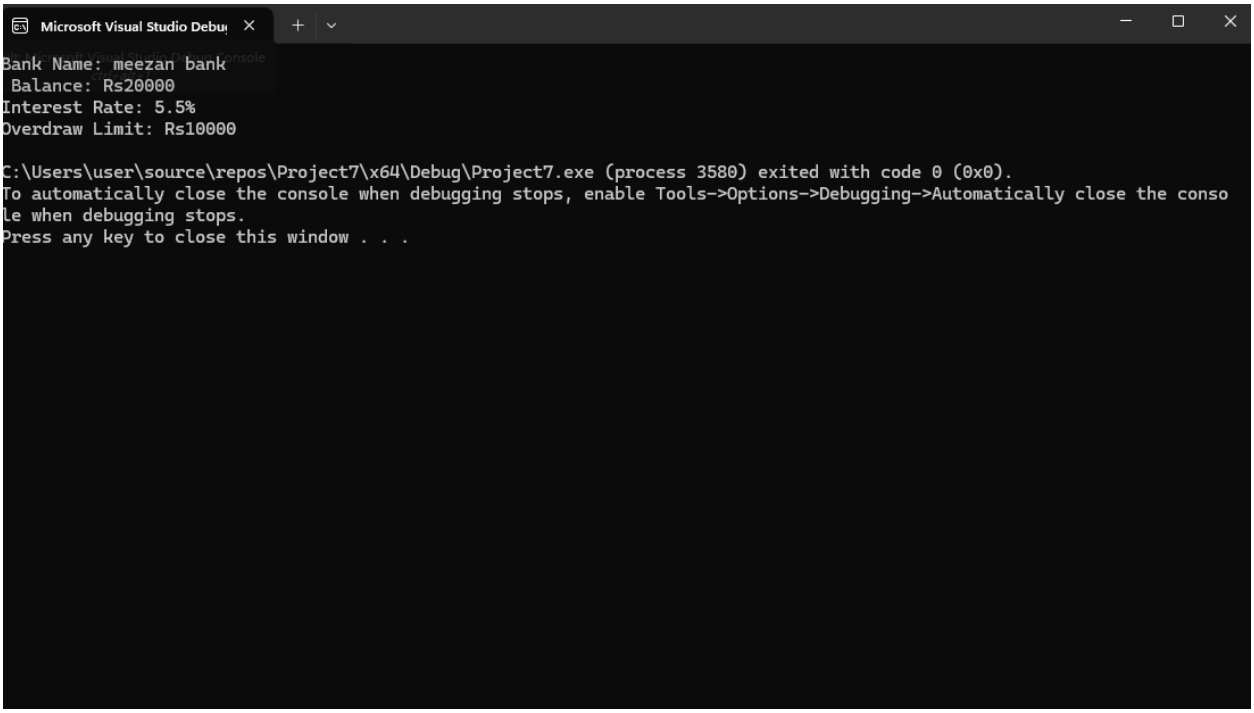
    void setBankDetails(string name, double bal, double rate, double limit) {
     bankName = name;
    balance = bal;
   setCheckingDetails(rate, limit);
    }

    void displayBankInfo() {
        cout << "Bank Name: " << bankName << endl;
        cout <<  " Balance: Rs" << balance << endl;
        displayCheckingInfo();
    }
};

int main() {
    BankAccount b;
    b.setBankDetails("meezan bank", 20000.00, 5.5, 10000.0);
    b.displayBankInfo();

    return 0;
}
```

Output:


Microsoft Visual Studio Debu
Bank Name: meezan bank
Balance: Rs20000
Interest Rate: 5.5%
Overdraw Limit: Rs10000
C:\Users\user\source\repos\Project7\x64\Debug\Project7.exe (process 3580) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .

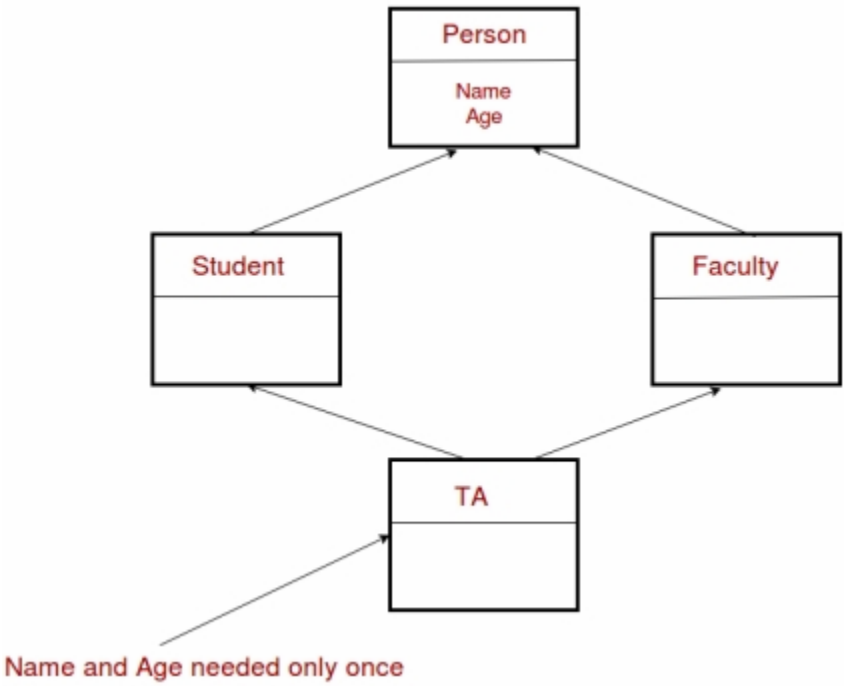
Task 3: Hybrid Inheritance (20 mark)

See the following class diagram and implement multiple inheritance relationship of classes in C++.

You need to implement following items for each class;

1. Constructors
2. Setters
3. Getters
4. Member Methods

Demonstrate working of your classes.



Program:

```
#include <iostream>
using namespace std;

class Person {
private:
    string name;
    int age;
```

```
public:
    Person() {
        name = "";
        age = 0;
    }

    void setPersonDetails(string n, int a) {
        name = n;
        age = a;
    }

    void displayPersonInfo() {
        cout << "Name: " << name << ", Age: " << age << endl;
    }
};

class Student : public Person {
private:
    string studentID;

public:
    Student() {
        studentID = "";
    }

    void setStudentDetails(string id) {
        studentID = id;
    }

    void displayStudentInfo() {
        cout << "Student ID: " << studentID << endl;
    }
```

```
};

class Faculty : public Person {
private:
    string facultyID;

public:
    Faculty() {
        facultyID = "";
    }

    void setFacultyDetails(string id) {
        facultyID = id;
    }

    void displayFacultyInfo() {
        cout << "Faculty ID: " << facultyID << endl;
    }
};

class TA : public Student, public Faculty {
private:
    string courseAssigned;

public:
    TA() {
        courseAssigned = "";
    }

    void setTADetails(string n, int a, string sid, string fid, string course) {
        Student::setPersonDetails(n, a);
        setStudentDetails(sid);
        setFacultyDetails(fid);
```

```
        courseAssigned = course;
    }

    void displayTAInfo() {
        Student::displayPersonInfo();
        displayStudentInfo();
        displayFacultyInfo();
        cout << "Course Assigned: " << courseAssigned << endl;
    }
};

int main() {
    TA ta;
    ta.setTADetails("Ali Raza", 19, "9864", "bscs", "cs");
    ta.displayTAInfo();
    return 0;
}
```

Output:

